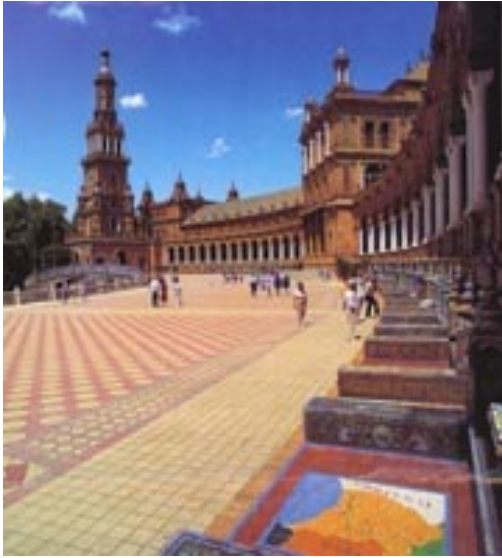


Effective Analytic Geometry Over Complete Fields

Special Session at the Joint AMS-RSME Meeting (Seville, June 18 & 20, 2003)



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<http://www.math.tamu.edu/~rojas/seville.pdf>

Invited Speakers: Vladimir Berkovich*, Andrei Gabrielov*, Thomas Hales, Maxim Kontsevich, Leonard Lipshitz*, Yuri Manin, Margarita Otero*, Luis-Miguel Pardo*, J. Maurice Rojas*, Thomas Scanlon, Lou van den Dries*, Daqing Wan.

(* =confirmed)

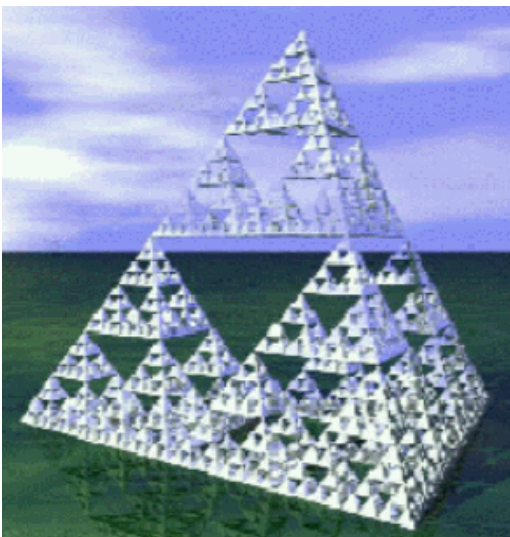
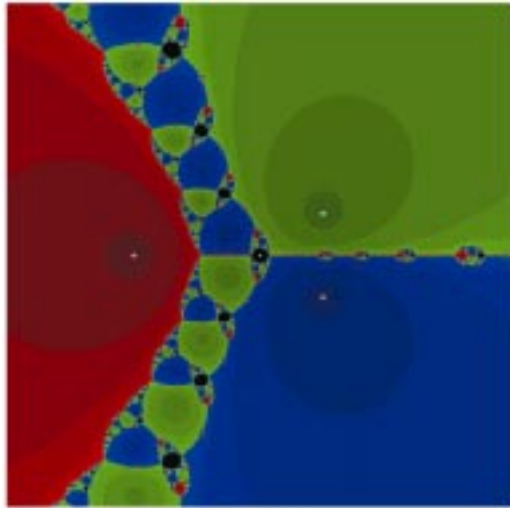
Striking recent advances in real algebraic geometry and arithmetic geometry reveal a deep connection between quantitative results over the real numbers and p -adic numbers. For instance, Khovanski's Theory of Fewnomials (from the late 1970's) gives bounds on the number of real solutions of sparse multivariate polynomial equations which are independent of the degree, and an even sharper analogous bound over arbitrary p -adic fields and number fields was proved by Rojas in 2001.

Since real algebraic geometry is central in applications in numerous areas of science and engineering, and since p -adic analysis is central in numerous areas within cryptography and computational number theory, a unified treatment would greatly benefit and unify many disparate groups of researchers and applied scientists.

The two primary goals of this special session are (a) to present the latest algorithmic, quantitative, and/or structural results from real and p -adic analytic geometry, and (b) to unify the researchers involved in these and other adjacent fields and to provide an "incubator" for future collaboration and growth.

Topics: Generalizations of Descartes' Rule to Higher Dimensions and Complete Fields (a.k.a. Fewnomial Theory Over Complete Fields), algorithms and effective estimates for cohomology of semi-algebraic and semi-Pfaffian sets, connections to semi-definite programming, cell decompositions (real and p -adic), quantifier elimination over the real and p -adic numbers, model-theoretic aspects (e.g., o -minimality and t -minimality), explicit complexity bounds, Lojasciewicz estimates (real, complex, and p -adic), the role of approximate computation over the real and p -adic numbers, amoebae of algebraic sets (complex and non-archimedean), tropical methods, Igusa zeta functions, motivic zeta functions, applications to mirror symmetry, as well as other approaches and applications.

$$F(x) = y^n$$



$$Z(f, s) := \int_{\mathbb{Z}_p^m} |f(x)|^s |dx| = \sum_{n \geq 0} \text{Vol}(Z_n) p^{-ns}$$